# BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

# PHYSICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

## Section 'A' (Multiple Choice Questions) (17)

**Maximum Marks:** 17

Note: Select the correct answer for each from the given options. (The 2025 paper prints these 17 items as a NOTE list with star-marked options; shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85 (Science Pre-Engineering & Pre-Medical Groups)

Qi. If charge on a particle moving in a magnetic field is doubled, while its velocity remains same, magnetic force on the particle will be:

(A) Doubled
(B) Halved
(C) Constant
(D) Quadrupled

Qii. According to Charle's law:

(A) VT =constant, when pressure is constant
(B) V/T =constant, when pressure is constant
(C) PV =constant, when temperature is constant
(D) V/P =constant, when temperature is constant

Qiii. The Primary purpose of a truth table in digital electronics is to:

(A) Determine input voltage
(B) Measure circuit resistance
(C) Calculate circuit power
(D) Analyze output states based on input states

Qiv. An example of an inertial frame of reference in our daily life is:

(A) Accelerating car
(B) Train moving with constant velocity
(C) Spinning platform
(D) Aeroplane encountering turbulence

Qv. The least number of energy levels required for LASER action are:

(A) Two
(B) Three
(C) Four
(D) Five

Qvi. If the temperature of a gas is tripled in Kelvin at constant volume, the pressure of the gas will be:

(A) Double
(B) Quadruple
(C) One third
(D) Triple

Qvii. A thermodynamics process at constant volume is called:

(A) Isothermal
(B) Isochoric
(C) Adiabatic
(D) Isobaric

Qviii. The net change in entropy of a system in a natural process is always:

(A) Positive
(B) Negative
(C) Zero
(D) Infinite

Qix. In purely resistive AC circuit, the phase difference between current and voltage is:

(A) 0°
(B) 45°
(C) 90°
(D) 180°

Qx. Property o a body by virtue of which, it tends to regain its original size and shape, when the applied force is removed this characteristics is called:

(A) Elasticity
(B) Rigidity
(C) Plasticity
(D) Compressibility

Qxi. An object is moving at a good of 0.9c and its mass is 1kg its relative mass will be:

(A) Infinite
(B) 1.2kg
(C) 2.3kg
(D) 1kg

Qxii. In an alternating current generator electrical energy is obtained by the conversion of:

(A) Wind energy
(B) Mechanical Energy
(C) Solar energy
(D) Chemical energy

Qxiii. Least amount of energy required to emit an electron from surface of metal, in photoelectric effect is called:

(A) Threshold frequency
(B) Binding energy
(C) Work function
(D) Stopping potential

Qxiv. To Produce X-rays, an element of high atomic weight is bombarded by:

(A) Photon
(B) Electrons
(C) Helium
(D) Neutron

Qxv. $zX^{A} \rightarrow z_{+1}Y^{A} + {}_{-1}e^{0}$, this equation presents:

(A) y − emission
(B) α − emission
(C) β − emission
(D) neutron − emission

Qxvi. In fission and fusion reaction energy is released due to:

(A) Chemical reaction
(B) Breaking of bond
(C) Conversion of matter in energy
(D) Combustion process

Qxvii. This derive is not used in diesel engine:

(A) Outlet valve
(B) Piston
(C) Sparking plug
(D) Injector

## Section 'B' (Short-Answer Questions) (36)

**Maximum Marks:** 36

> Note: Answer any NINE part questions from this section. All questions carry equal marks. Draw diagrams where necessary.

Q2.

(i) The $_{15}{}^{32}P$ isotopes of phosphorus has a half-life of 14.28 days. Find is its decay constant.

(ii) State the laws governing the black body radiations. Also write their mathematical expressions.

(iii) Calculate the wavelength of X-rays, if energy of a photon emitted by the X-rays machine is $1.9878 \times 10^{-13}$ J.

(iv) Draw the circuit symbol of OR and AND gates and write Truth Tables for two inputs.

(v) In an RLC circuit, the resistance (R) is 50 ohms, the inductan . (L) is 0.1 H and the capacitance (C) is 50μF. calculate its resonance frequency.

(vi) A galvanometer has a resistance of 50Ω and it giv full scale deflection at 4 mA. A voltmeter is made using this galv nometer by connecting a 5kΩ resistor in series. Calculate the maximum voltage that can be measured using this voltmeter.

(vii) A system undergoes a thermodynamic process where the pressure is kept constant at 15kPa. If the volume increases from $0.05m^3$ to $0.08m^3$ and the change in internal energy is 150 J. Calculate the heat added to the system.

(viii) What is the name of the EMF induced in a straight conductor moving with uniform velocity in a magnetic field?

(ix) Derive its mathematical expression $C_p - C_v = R$.

(x) State law of radioactive decay and write its mathematical expression.

(xi) How much heat will be required to raise the temperature of 1 kg of lead from 30°C to 105 °C? (specific heat of lead = 0.128J/kg°C)

(xii) Define magnetic flux. Under what condition the magnetic flux will be maximum?

(xiii) Prove that the speed of a moving rod at which its relative length is equal to half of its proper leugth is √3c/2.

(xiv) What is meant by LASER? Define metastable state OR population inversion.

## Section "C" (Detailed Answer Questions) (32)

**Maximum Marks:** 32

> Note: Attempt any TWO questions from this section. All questions carry equal marks. Draw diagrams where necessary.

Q3.

(a) State the basic assumptions of kinetic molecular theory of gases. Derive the equation for pressure of an ideal gas, where $P = 1/3\ \rho\bar{v}^2$ and ρ represents the density of the gas.

(b) State Ampere's law. Derive an expression for the magnetic field of induction 'B' due to current passing through a long Solenoid OR Toroid.

Q4.

(a) What is photoelectric effect? Define the following terminologies *work function *stopping potential *threshold frequency and derive Einstein's photoelectric equation.

(b) What is Carnot Engine? Describe the process of a Carnot cycle and derive mathematical expression for its efficiency.

Q5.

(a) Write the postulates of special theory of relativity. Explain any two of its results.

(b) What is a mass spectrograph? Explain its construction, principle and working. Derive its relevant mathematical expression.